

Microprocessor

Applications:

Gaming

Web browsing

Photo editing

Creating documents

Mathematical Calculation

Simulations



Microcontroller

They are designed to perform specific task



Microcontroller

Applications:

Digital Camera



Washing Machines

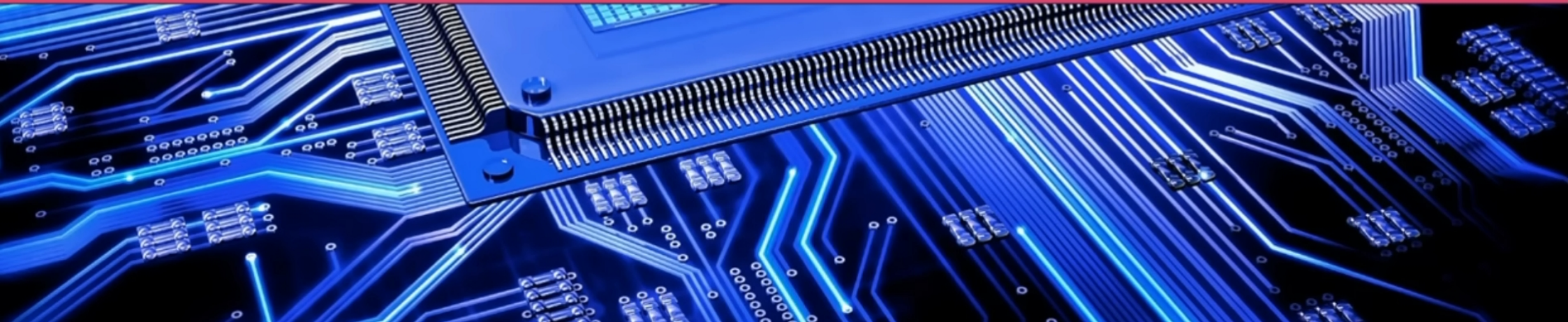


Microwave Oven

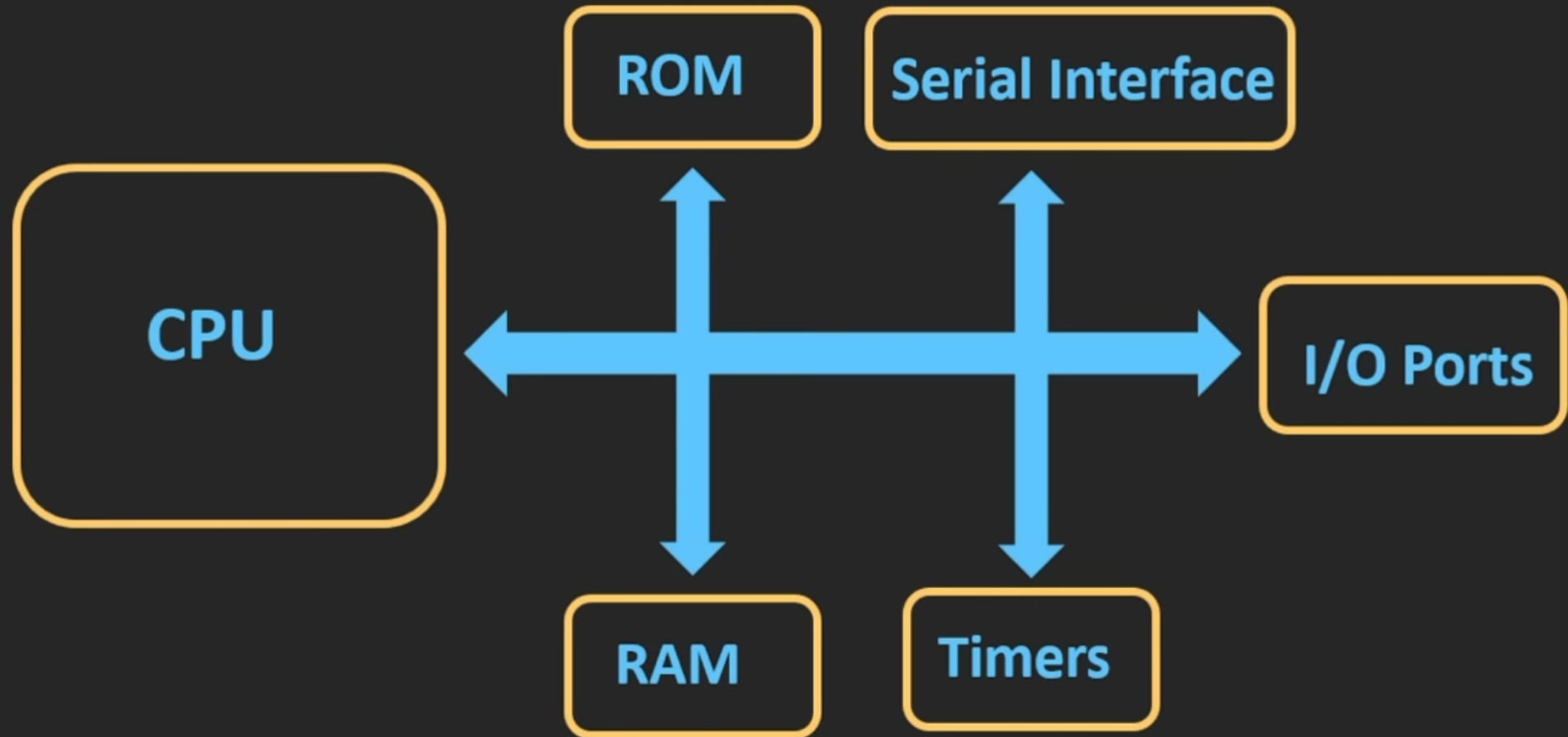




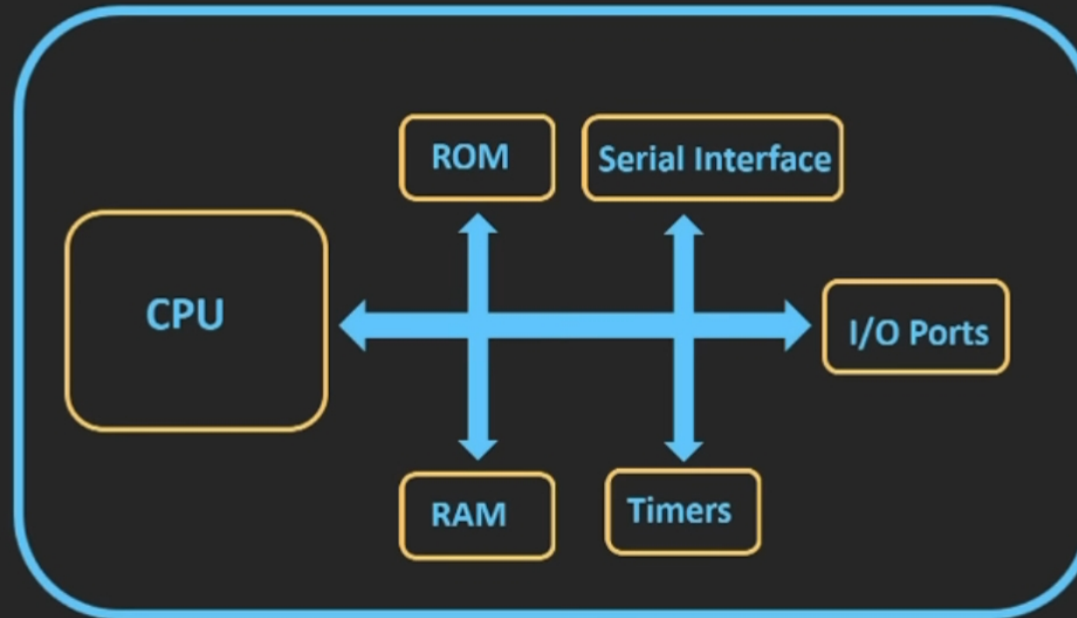
Internal Structure



Microcontroller



Microcontroller



Processing Power and Memory

Microprocessor

Clock Speed: 1 GHz to 4 GHz

RAM: 512 MB to up to 32 GB

**ROM
(Hard Disk):** 128 GB to up to 2 TB

Peripheral USB

Interfaces: High Speed Ethernet
UART

Microcontroller

Clock Speed: 1 MHz up to 300 MHz

RAM: 2 KB to up to 256 KB

Flash Memory: 32 KB to up to 2 MB

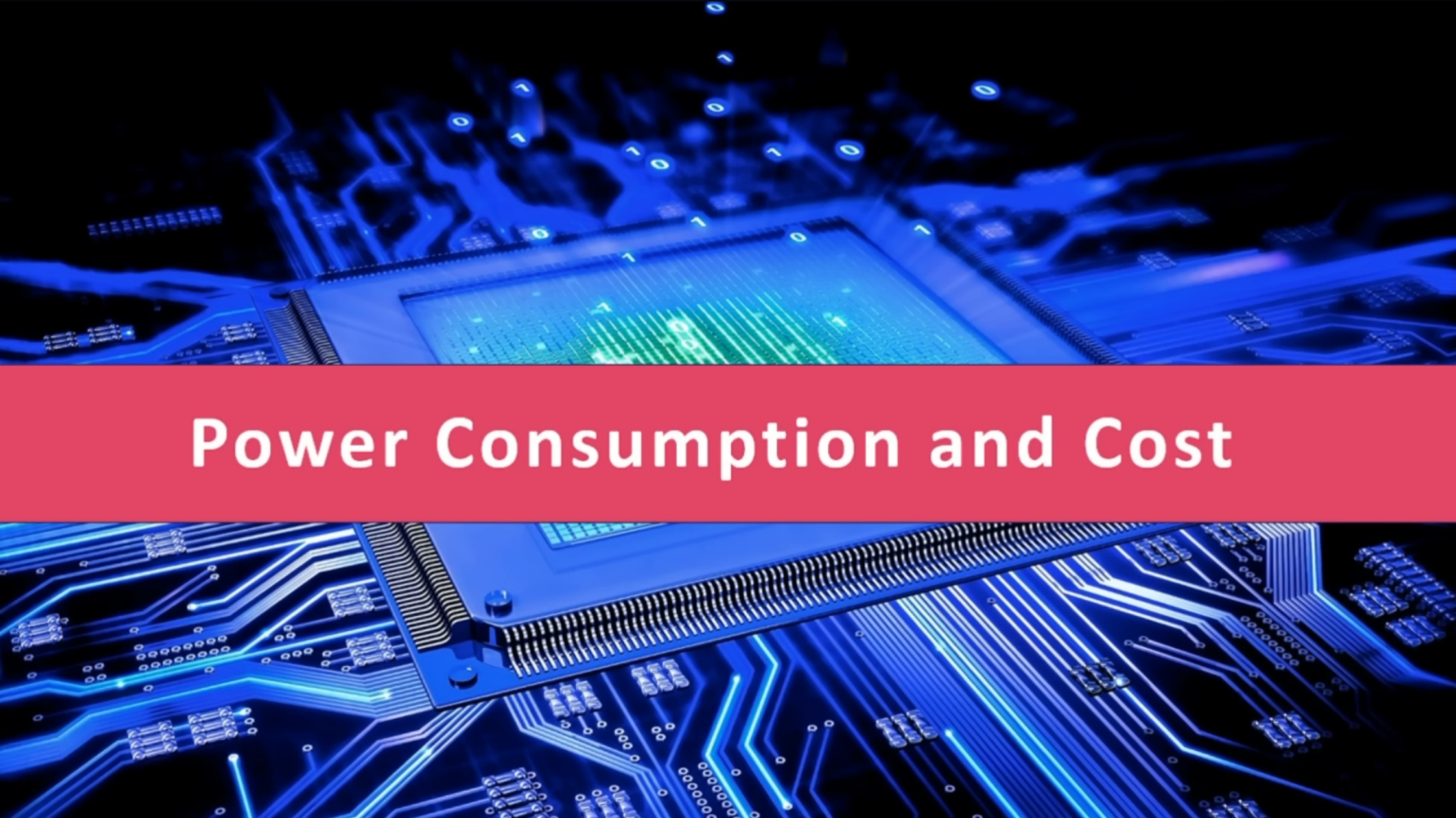
Peripheral I2C

Interfaces: SPI

UART

Microprocessor: 32 bit and 64 bit

Microcontroller: 8 bit, 16 bit and 32 bit



Power Consumption and Cost

Microprocessor

Power Consumption: High

Cost: High

Summery

Parameters	Microprocessors	Microcontroller
Applications	Gaming, web browsing, document creation etc.	Dedicated for specific tasks (Camera, washing machine etc.)
Internal Structure	Memory and I/O devices connected externally	CPU, memory and I/O are present internally
Cost	High	Low
Power Consumption	High	Low
Memory (RAM)	512 MB to 32 GB	2KB to up to 256 KB
Storage	Hard Disk (128 GB to up to 2 TB)	Flash memory (32 KB to 2 MB)
Peripheral Interfaces	USB, UART, High speed Ethernet	UART, I2C, SPI